

Muad Abd El Hay

Curriculum vitae

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Nationality: German, Israeli/Palestinian

Born on 15.11.1989 in Kfar Sava, Israel



Education

- 2022–Now **PostDoc**, *ZeroNoise Lab*, Ernst Struengmann Institute, Frankfurt am Main - Germany.
Core funding of Max Planck junior research group
- 2021–2022 **PostDoc**, *AG Siemens*, Institute of Pharmacology, Heidelberg - Germany.
- 2015–2021 **Ph.D.**, *Ruperto Carola University*, Heidelberg - Germany, *Magna cum Laude*.
- 2011–2014 **M.Sc.**, *Ulm University*, Ulm - Germany, 1.2.
Molecular Medicine
- 2008–2011 **B.Sc.**, *Ulm University*, Ulm - Germany, 2.3.
Molecular Medicine
- 2005–2007 **Bagrut**, *Amal 1 High School*, Tira - Israel, 97%.
with focus on Biotechnology, Biochemistry, and Biology

Theses

- Ph.D. *The roles of TRPV1 and TRPM2 in warm temperature detection*
Supervisors Prof. Dr. Jan-Erik Siemens and Jun. Prof. Dr. Daniela Mauceri
- Master's *The effect of TRAIL on adipokine secretion in primary and SGBS adipocytes*
Supervisors PD Dr. Pamela Fischer-Posovszky and MD Dr. Jochen Lennerz
- Bachelor's *Molecular characterization of *S. agalactiae* strains causing streptococcal toxic shock syndrome*
Supervisors Prof. Dr. Barbara Spellerberg and Prof. Dr. Holger Barth

Publications

- [1] **Abd El Hay, Muad**. The role of TRPV1 and TRPM2 in warm temperature detection. <https://archiv.ub.uni-heidelberg.de/volltextserver/30387/>, 2022.
- [2] K. Schrenk-Siemens, J. Pohle, C. Rostock, **Abd El Hay, M.**, R. M. Lam, M. Szczot, S. Lu, A. T. Chesler, and J. Siemens. Human Stem Cell-Derived TRPV1-Positive Sensory Neurons: A New Tool to Study Mechanisms of Sensitization. *Cells*, 11(18):2905, Sept. 2022.
- [3] G. B. Kamm, J. C. Boffi, K. Zuza, S. Nencini, J. Campos, K. Schrenk-Siemens, I. Sonntag, B. Kabaoğlu, **Abd El Hay, M.**, Y. Schwarz, A. Tappe-Theodor, D. Bruns,

- C. Acuna, T. Kuner, and J. Siemens. A synaptic temperature sensor for body cooling. *Neuron*, 109(20):3283–3297.e11, Oct. 2021.
- [4] V. Zoller, J.-B. Funcke, J. Roos, M. Dahlhaus, **Abd El Hay, M.**, K. Holzmann, R. Marienfeld, T. Kietzmann, K.-M. Debatin, M. Wabitsch, and P. Fischer-Posovszky. Trail (TNF-related apoptosis-inducing ligand) induces an inflammatory response in human adipocytes. *Sci Rep*, 7(1):5691, Dec. 2017.
 - [5] P. Sendi, **Abd el Hay, M.**, C. M. Brandt, and B. Spellerberg. Group B Streptococcal Toxic Shock Syndrome and covR/S Mutations Revisited. *Emerg Infect Dis*, 23(1):150–152, Jan. 2017.
 - [6] V. Zoller, J.-B. Funcke, M. Keuper, **Abd El Hay, M.**, K.-M. Debatin, M. Wabitsch, and P. Fischer-Posovszky. TRAIL (TNF-related apoptosis-inducing ligand) inhibits human adipocyte differentiation via caspase-mediated downregulation of adipogenic transcription factors. *Cell Death Dis*, 7(10):e2412, Oct. 2016.
 - [7] J.-B. Funcke, V. Zoller, **Abd El Hay, M.**, K.-M. Debatin, M. Wabitsch, and P. Fischer-Posovszky. TNF-related apoptosis-inducing ligand promotes human preadipocyte proliferation via ERK1/2 activation. *The FASEB Journal*, Apr. 2015.
 - [8] J. Lemke, S. von Karstedt, **Abd El Hay, M.**, A. Conti, F. Arce, A. Montinaro, K. Papenfuss, M. A. El-Bahrawy, and H. Walczak. Selective CDK9 inhibition overcomes TRAIL resistance by concomitant suppression of cFlip and Mcl-1. *Cell Death Differ*, 21(3):491–502, Mar. 2014.

Experience

Internships

- 10/12 – 04/13 **UCL Cancer Institute**, University College London, With a stipend from the BCCG foundation and ERASMUS SMP.
 Prof. Dr. Henning Walczak
 Investigating the effect of a CDK9 inhibitor on the TRAIL sensitivity of NSCLC

Conferences

- 09/23 **INSIGHT - Integrating Science and Practice of Psychedelic States**, Berlin, Germany.
- 07/22 **FENS2022 - Federation of European Neuroscience Societies Forum**, Paris.
 Poster presentation and abstract submission
- 09/21 **INSIGHT - Integrating Science and Practice of Psychedelic States**, Berlin, Germany.
- 07/20 **FENS2020 - Federation of European Neuroscience Societies Forum**, Virtual.
 Poster presentation and abstract submission
- 09/19 **INSIGHT - Integrating Science and Practice of Psychedelic States**, Berlin, Germany.
- 07/18 **FENS2018 - Federation of European Neuroscience Societies Forum**, Berlin.
 Poster presentation and abstract submission
- 09/16 **Beyond psychedelics - Multidisciplinary conference on psychedelic research**, Prague, Czech Republic.
- 04/14 **Breaking Convention - Multidisciplinary conference on psychedelic research**, London, United Kingdom.

- 06/17 **SFB1185 - Neurocircuits symposium**, Heidelberg, Germany.
Poster presentation, abstract submission, and poster prize
- 09/11 **63th Annual Conference of the German Society for Hygiene and Microbiology**, Essen, Germany.
Poster presentation and abstract submission

Work Experience

- 01/22 – 03/22 **Scientific review of MOLECULES - Augmented Psychotherapy Training (APT)**, MIND Foundation, Berlin - Germany, [link](#).
- 08/09 – 03/14 **Corona Web Administrator**, Medical Faculty, University Hospital Ulm.
Prof. Dr. Thomas Wirth – Dr. Hubert Liebhardt
- 03 – 04/10 **Tutor for Organic Chemistry**, Department of Organic Chemistry, Ulm.
Prof. Dr. Hans-Ullrich Siehl

Extra Curricular Activities

- 2016 – Now **Initiator and Coordinator**, uniMIND Project, MIND Foundation, Journal club and discussion group about research and therapeutic use of psychedelic substances.
[link](#)
- 2016 – 2022 **Coordinator**, uniMIND Heidelberg, Bi-weekly journal club about psychedelics in research. About half of the talk/presentations were given by me.
- 2009 – 2013 **Organization and Moderation**, 2nd, 3rd, 5th and 6th Annual "Tag der Molekularen Medizin" students symposium, University of Ulm.
- 04/11 – 03/14 **Board Member**, Biotechnologic Student's Initiative (btS), University of Ulm.
- 10/08 – 03/14 **Active Member**, Student's Council, University of Ulm.

Outreach

- 03/23 **Workshop**, Drug FAQs - Drug harm-reduction workshop, Kabinett der Kuriositaeten, Frankfurt - Germany.
- 09/22 **Workshop**, Drug FAQs - Drug harm-reduction workshop (in Arabic), Al-Qarya project - Israel, Online.
- 09/22 **Interview**, To Hallucinate or Not: The Big Questions Facing Psychedelics at FENS 2022, Technology Networks, [link](#).
- 07/21 **Project**, Care Bears - Drug-checking and harm-reduction stand, Kiezburn Festival, Brandenburg - Germany.
- 12/20 **Podcast**, Can you Feel the Heat? - Temperature Perception and the Problems of Modern Science, The Transformative Ideas Podcast, [link](#).
- 12/20 **Podcast**, Challenges and Opportunities of Psychedelic Research, The Transformative Ideas Podcast, [link](#).
- 03/17 **Blogpost**, A lid for every pot - Crystal structure of LSD-bound serotonin receptor, MIND Foundation blog, [link](#).
- 2014 – 2018 **Project**, Independent drug-checking and harm-reduction activities, Midburn Festival, Negev - Israel.

Methods and Skills

- Basics Cell culture, bacterial culture, neuronal culture preparation

Genomics	DNA- and RNA-extraction, q/PCR, cloning, sequencing, shRNA, whole-mount ISH
Proteomics	Protein extraction, western blot, ELISA, protein purification
FACS	Basics, cell cycle analysis, apoptosis detection
Imaging	Fluorescence microscopy, calcium imaging, voltage imaging
Animal work	FELASA B certificate, hargreaves assay, two-place preference test, CCI-surgery, markerless pose estimation
Misc.	Zebrafish breeding, microinjection, Lenti- and AAV-virus production, electrophysiology

Computer skills and software packages

R	Expert	Python	Advanced
DeepLabCut	Expert	Suite2p	Expert
Lightning Pose	Expert	Bash	Advanced
Emacs	Advanced	Inkscape/Gimp	Advanced
		LaTeX	Advanced

Languages

Arabic	Native
German	Fluent
English	Fluent
Hebrew	Fluent
Polish	Basic